SUPRIYA GHODKE

Senior Hardware Engineer
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Professional Summary

Senior Embedded Hardware Engineer with 6.8+ years of experience designing and delivering production-grade embedded systems across IoT, industrial, and automotive domains. Expertise in multilayer PCB design, power electronics, mixed-signal systems, and system-level architecture.
Strong track record of owning end-to-end hardware development and collaborating across cross-functional teams, with increasing responsibility in design reviews, system-level decisions, and mentoring — positioning toward **Hardware Team Lead and system ownership roles**.

Key Impact Highlights

- Reduced system power consumption by **25–30%** through optimized power architecture
- Improved signal accuracy by **~20%** via advanced analog front-end design
- Delivered production-ready hardware supporting **10K+ unit deployments**
- Achieved **~15% BOM cost reduction** through component optimization
- Accelerated hardware bring-up and debugging cycles by **~30%**

Core Technical Skills

Hardware Design & Architecture
- Embedded Hardware Design, System Architecture, Mixed-Signal Systems
- Multilayer PCB Design (Altium, OrCAD, Cadence)
- Signal Integrity, EMI/EMC Design

Power Electronics
- AC-DC Flyback, DC-DC Converters, LDO
- Battery Systems, Power Optimization

Embedded Systems
- STM32, Silicon Labs SoC
- SPI, I2C, UART, CAN, RS485, BLE, NFC

Validation & Engineering
- Hardware Bring-up, Debugging & Validation
- SPICE Simulation (LTSpice, PSpice)
- DFM / DFT / DFA, FMEA, DFMA
- Component Selection & BOM Optimization

Professional Experience

Senior Hardware Engineer
Ceinsys Tech Ltd | Dec 2024 – Present
- Led end-to-end hardware development lifecycle from concept to production for embedded systems
- Reduced system power consumption by **25–30%** through optimized power architecture and component selection
- Designed and validated DC-DC power systems ensuring stability under varying load conditions
- Developed analog and mixed-signal circuits improving signal performance and system reliability
- Accelerated hardware bring-up and debugging cycles by **~30%** using structured validation approaches
- Ensured EMI/EMC-compliant designs aligned with industry standards (Surge, EFT, ESD, CE)
- Reviewed multilayer PCB layouts focusing on signal integrity, power integrity, and manufacturability
- Mentored junior engineers and contributed to structured design reviews

Senior Hardware Engineer
L&T Technology Services | Jan 2022 – Oct 2024
- Designed hardware schematics for automotive ECUs and cluster systems using Cadence
- Improved signal accuracy by **~20%** through optimized analog front-end design and noise reduction techniques
- Executed hardware validation, debugging, and performance optimization
- Conducted simulation-based validation using PSpice to ensure design robustness
- Collaborated with global clients including Intel, Cisco, and Marelli (Ferrari, Nissan, Mitsubishi)
- Optimized component selection aligned with automotive-grade standards

Embedded Hardware Engineer
Robert Bosch | Jan 2019 – Mar 2021

- Defined system-level hardware architecture for IoT and energy management systems
- Designed precision sensing circuits using ADC and signal conditioning techniques
- Developed power supply circuitry for embedded platforms
- Applied DFMA and FMEA methodologies for reliability-driven design
- Managed component selection, validation, and technical documentation

IoT Engineer
Stanley Black & Decker | Jul 2018 – Jan 2019

- Developed embedded systems using STM32 and sensor interfaces
- Designed current sensor PCB using Altium Designer
- Performed system-level validation and testing
- Collaborated with suppliers for component selection and design alignment

Key Projects

Connected Water Meter (IoT Device)

- Designed battery-powered embedded hardware with Sub-GHz communication for long-range connectivity
- Delivered production-ready system supporting **10K+ deployments**
- Developed precision sensing system using Wheatstone bridge and high-precision ADC
- Implemented low-noise signal conditioning improving measurement accuracy
- Optimized system for low power consumption enabling extended battery life

Energy Management System (Commercial Refrigeration)

- Designed AC-powered system using flyback converter architecture
- Developed energy monitoring and load control circuits
- Integrated STM32-based control system with current sensing
- Enabled BLE and NFC connectivity for configuration and communication

Battery Management System (BMS) Redesign

- Improved PCB layout for signal integrity and robustness
- Optimized power paths and thermal performance
- Enhanced system reliability and electrical stability

Motion Monitoring System (BLE)

- Developed accelerometer-based motion sensing system
- Implemented real-time data transmission over BLE
- Designed threshold-based alert mechanism

Education & Certifications

Post Graduate Program – Artificial Intelligence & Machine Learning
University of Texas at Austin | 2020 – 2021
B.E. – Instrumentation & Control
Cummins College of Engineering, Pune | 2015 – 2018
Diploma – Electronics & Communication
Government Polytechnic, Beed | 2012 – 2015
Internal Auditor Quality Management Systems (ISO 9001:2015) Link
Ascent Academy (CS Tech AI) | Nov '25